

Topic 8: Individual Demand

(Loosely based on Ch. 5 From the
book)



① Utility Maximization Problem of the Consumer

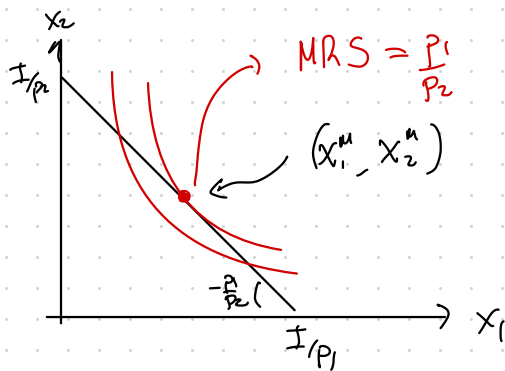
$$\max_{(x_1, x_2)} U(x_1, x_2)$$

$$(x_1, x_2)$$

$$\text{s.t. } p_1 x_1 + p_2 x_2 \leq I$$

Choice variables: x_1, x_2 endogenous variables

Parameters (exogenous variables): p_1, p_2, I



$$\left. \begin{array}{l} x_1^M(p_1, p_2, I) \\ x_2^M(p_1, p_2, I) \end{array} \right\} \begin{array}{l} \text{System of } \underline{\text{Marshallian}} \\ \underline{\text{demands}} \text{ for the consumer} \\ \text{(functions of prices and} \\ \text{income)} \end{array}$$

② Expenditure Minimization problem:

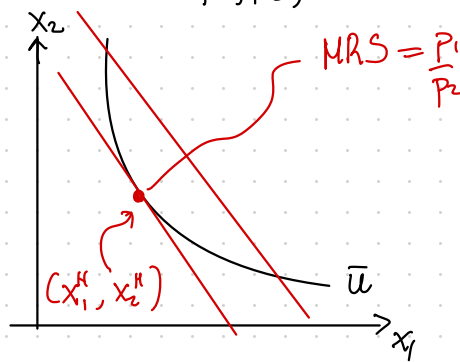
$$\min_{(x_1, x_2)} p_1 x_1 + p_2 x_2$$

$$(x_1, x_2)$$

$$\text{s.t. } U(x_1, x_2) \geq \bar{U}$$

Choice variables: x_1, x_2

Parameters: $p_1, p_2, \bar{U}$



$$\left. \begin{array}{l} x_1^H(p_1, p_2, \bar{U}) \\ x_2^H(p_1, p_2, \bar{U}) \end{array} \right\} \begin{array}{l} \text{System of } \underline{\text{Hicksian}} \underline{\text{demands}} \\ \text{for the consumer} \\ \text{(functions of prices and utility)} \end{array}$$

Examples:

① $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ $0 < \alpha < 1$ is a parameter Cobb-Douglas Utility

$$MRS = \frac{\alpha}{1-\alpha} \frac{x_2}{x_1}$$

$$\max_{(x_1, x_2)} x_1^\alpha x_2^{1-\alpha}$$

$$\text{s.t. } p_1 x_1 + p_2 x_2 = I$$

$$\text{Optimality condition: } \frac{\alpha}{1-\alpha} \frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow x_2 = \frac{1-\alpha}{\alpha} \frac{p_1}{p_2} x_1$$

Plug into the constraint (Budget Line)

$$p_1 x_1 + p_2 \left(\frac{1-\alpha}{\alpha} \frac{p_1}{p_2} x_1 \right) = I$$

$$\left(p_1 + \frac{1-\alpha}{\alpha} p_1 \right) x_1 = I$$

$$\left(1 + \frac{1-\alpha}{\alpha} \right) p_1 x_1 = I$$

$$\left(\frac{\alpha + 1 - \alpha}{\alpha} \right) p_1 x_1 = I$$

$$x_1^M(p_1, p_2, I) = \frac{\alpha I}{p_1}$$

$$x_2^M(p_1, p_2, I) = \frac{(1-\alpha) I}{p_2}$$

Total optimal expenditure on x_1 : $x_1^M p_1 = \alpha I$
" " " " x_2 : $x_2^M p_2 = (1-\alpha) I$

$$x_1^M(p_1, p_2, I) = \frac{\alpha I}{p_1}$$

$$x_2^M(p_1, p_2, I) = \frac{(1-\alpha)I}{p_2}$$

System of Marshallian demands for the Cobb-Douglas Utility.

"Demand Curves"

For x_1 : fix $\bar{p}_2$, $\bar{I}$, and allow p_1 to change:

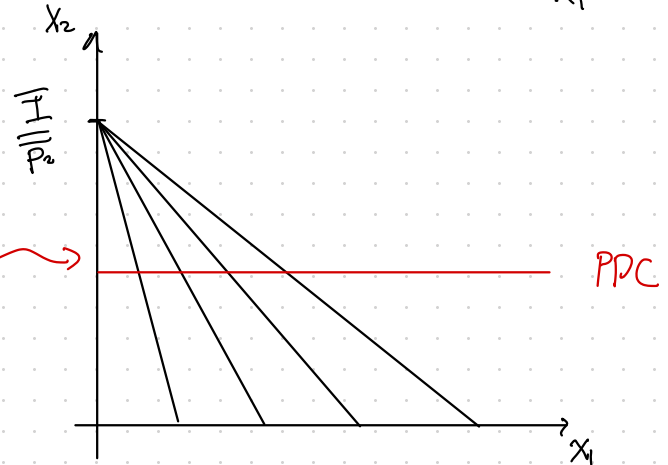
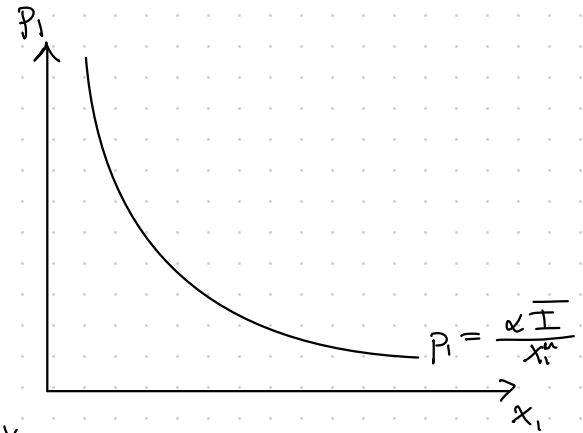
$$x_1^M(p_1) = \frac{\alpha \bar{I}}{p_1} \rightarrow p_1 = \frac{\alpha \bar{I}}{x_1^M}$$

Inverse demand

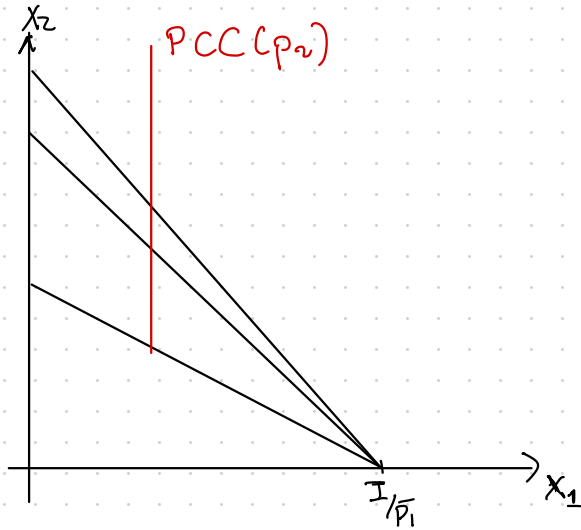
$$x_2^M(p_1) = \frac{(1-\alpha)\bar{I}}{\bar{p}_2}$$

Price Consumption Curve:

the set of optimal bundles when allowing one of the prices to change



$$PPC(p_2) = \left(\frac{\alpha \bar{I}}{\bar{p}_1}, \frac{(1-\alpha) \bar{I}}{p_2} \right)$$



Consider the Marshallian Demand System

$$x_1^m(p_1, p_2, I), x_2^m(p_1, p_2, I)$$

→ If we change p_1 , while holding everything else fixed

- $(x_1, p_1) : x_1^m(p_1, \bar{p}_2, \bar{I})$ demand curve

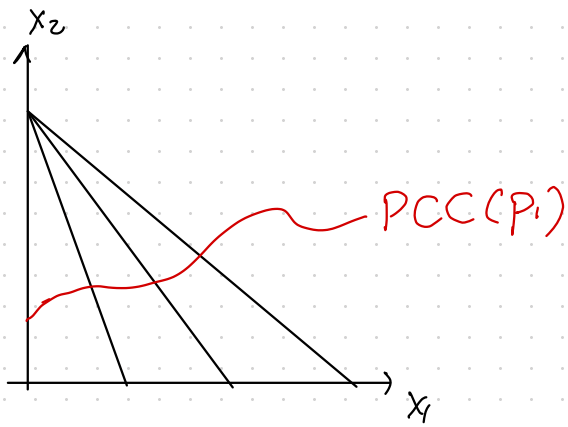
- In (x_1, x_2) "Consumption space"

$$PCC(p_1) = (x_1^m(p_1, \bar{p}_2, \bar{I}), x_2^m(p_1, \bar{p}_2, \bar{I}))$$

→ If we change p_2 , holding $\bar{p}_1, \bar{I}$ fixed

- $(x_2, p_2) : x_2^m(\bar{p}_1, p_2, \bar{I})$ demand curve for good 2

- In $(x_1, x_2) : PCC(p_2) = (x_1^m(\bar{p}_1, p_2, \bar{I}), x_2^m(\bar{p}_1, p_2, \bar{I}))$



Slope of the PCC is related to the own price and cross-price elasticities of demand.

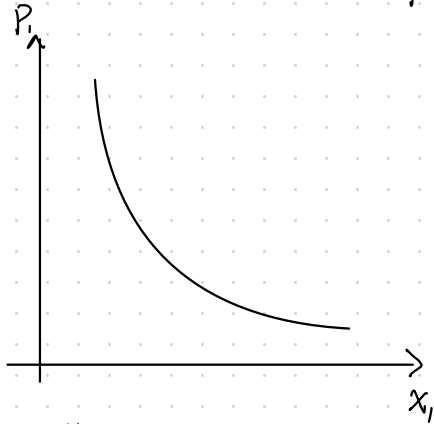
Example: Perfect Complements

$$u(x_1, x_2) = \min\left(\frac{x_1}{\alpha}, \frac{x_2}{\beta}\right)$$

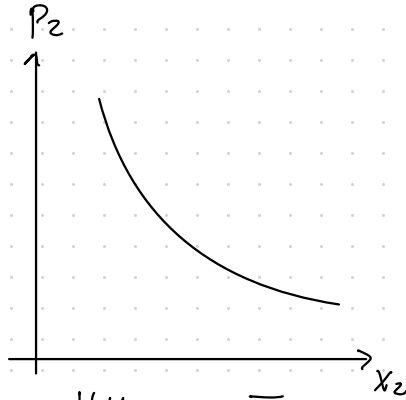
$$x_1^m(p_1, p_2, I) = \frac{\alpha I}{\alpha p_1 + \beta p_2}$$

$$x_2^m(p_1, p_2, I) = \frac{\beta I}{\alpha p_1 + \beta p_2}$$

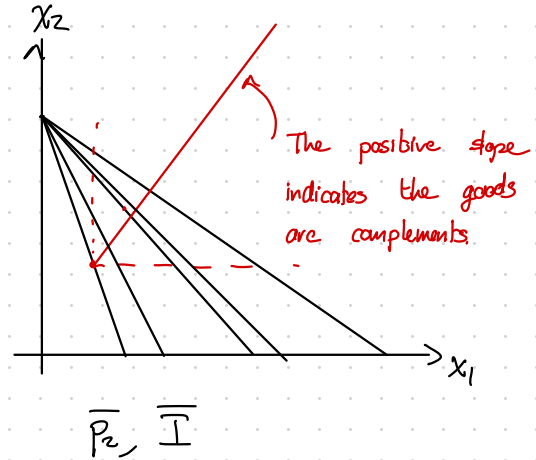
$$PCC(p_1) = \left(\frac{\alpha \bar{I}}{\alpha \bar{p}_1 + \beta \bar{p}_2}, \frac{\beta \bar{I}}{\alpha \bar{p}_1 + \beta \bar{p}_2} \right)$$



Holding p_2, I



Holding $\bar{p}_1, \bar{I}$



$\bar{p}_2, \bar{I}$

Example : CES utility ($\delta = 1/2$)

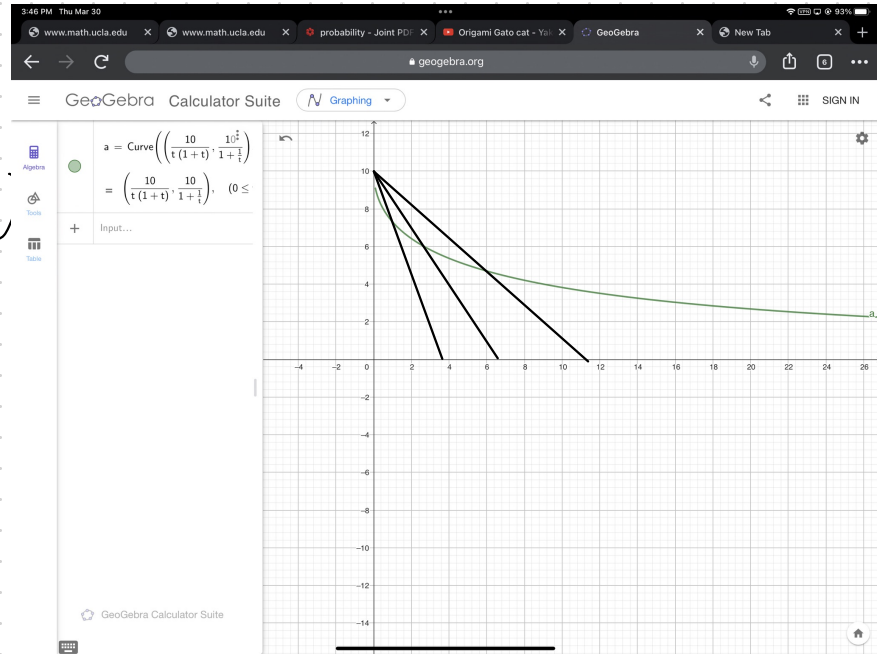
$$u(x_1, x_2) = x_1^{1/2} + x_2^{1/2} = \sqrt{x_1} + \sqrt{x_2}$$

$$MRS = \frac{x_1^{-1/2}}{x_2^{-1/2}} = \left(\frac{x_2}{x_1}\right)^{1/2}$$

$$x_1^m = \frac{I}{p_1 \left(1 + \frac{p_1}{p_2}\right)}$$

$$x_2^m = \frac{I}{p_2 \left(1 + \frac{p_2}{p_1}\right)}$$

$$PCC(p_1) = \left(\frac{10}{p_1(1+p_1)}, \frac{10}{1 \cdot (1+\frac{1}{p_1})} \right)$$
$$\bar{I} = 10$$
$$\bar{p}_2 = 1$$



Relationship between PPC & Elasticity of Demand.

$$\epsilon_{x_1, p_1} = \frac{\Delta \% x_1}{\Delta \% p_1}$$

$$\epsilon_{x_2, p_1} = \frac{\Delta \% x_2}{\Delta \% p_1}$$

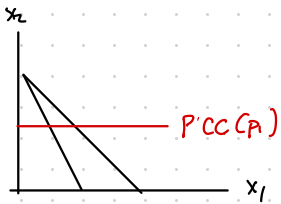
The share spent on good i

$$S_i = \frac{p_i x_i}{I}$$

$$S_1 \epsilon_{x_1, p_1} + S_2 \epsilon_{x_2, p_1} = -S_1 \Delta p_1$$

Cournot Aggregation

$$S_1 \epsilon_{x_1, p_1} + S_2 \epsilon_{x_2, p_1} = -S_1 \Delta p_1$$

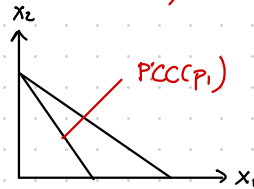


$$\Delta \% x_1 > 0, \quad \Delta \% x_2 = 0$$

$$\epsilon_{x_2, p_1} = 0$$

$$S_1 \epsilon_{x_1, p_1} + S_2 \epsilon_{x_2, p_1} = -S_1$$

$$\Rightarrow \epsilon_{x_1, p_1} = -1$$



$$\Delta \% p_1 < 0$$

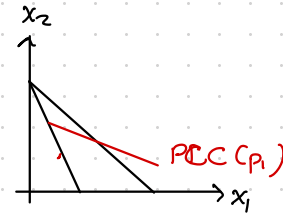
$$\Delta \% x_1 > 0, \quad \Delta \% x_2 > 0, \quad \Delta \% p_1 < 0$$

$$S_1 \epsilon_{x_1, p_1} + S_2 \epsilon_{x_2, p_1} = -S_1$$

$$\epsilon_{x_1, p_1} + \frac{S_2}{S_1} \epsilon_{x_2, p_1} = -1$$

$$\epsilon_{x_1, p_1} = -1 - \frac{S_2}{S_1} \epsilon_{x_2, p_1} > -1$$

$$\Rightarrow |\epsilon_{x_1, p_1}| < 1$$



$$\Delta \% p_1 < 0$$

$$\epsilon_{x_1, p_1} < 0, \quad \epsilon_{x_2, p_1} > 0$$

$$\epsilon_{x_1, p_1} = -1 - \underbrace{\frac{S_2}{S_1} \epsilon_{x_2, p_1}}_{< 0} < -1$$

$$\Rightarrow |\epsilon_{x_1, p_1}| > 1$$

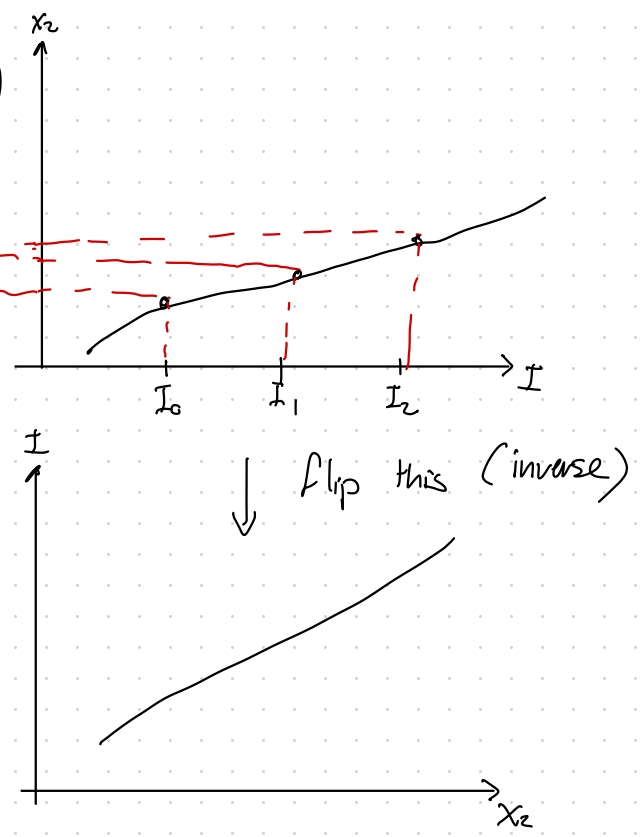
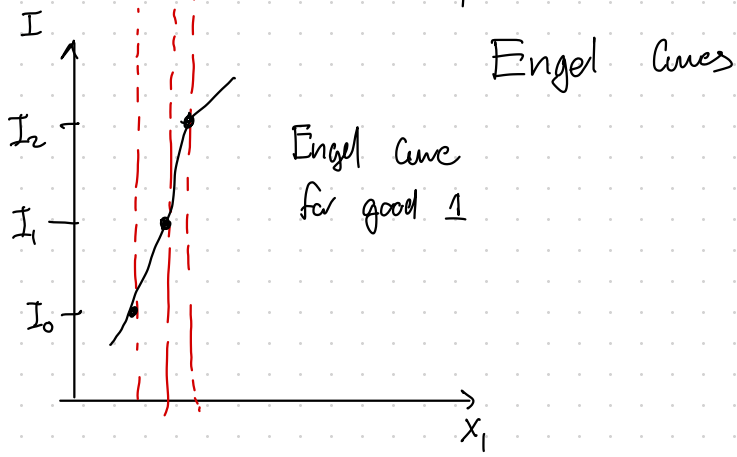
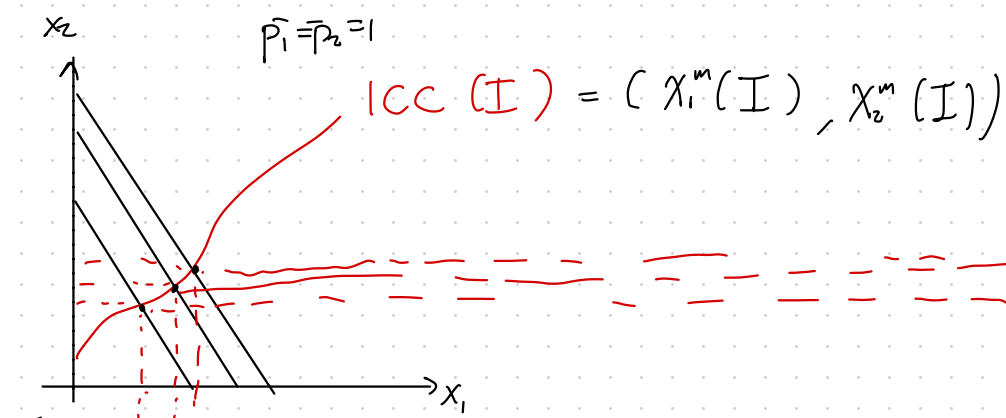
Consider again $x_1^m(p_1, p_2, I)$
 $x_2^m(p_1, p_2, I)$

Income Consumption Curve (ICC) : optimal set of bundles when holding p_1, p_2 fixed and changing Income.

$$ICC(I) = (x_1^m(\bar{p}_1, \bar{p}_2, I), x_2^m(\bar{p}_1, \bar{p}_2, I))$$

↳ (x_1, x_2) plotted in consumption space.

Engel Curves : trace optimal choices of each good as income changes in (x_1, I) , & (x_2, I)



Income Elasticity: $\epsilon_{i,I} = \frac{\Delta\% X_i}{\Delta\% I}$

Ranges for $\epsilon_{i,I}$	Good i is:
$\epsilon_{i,I} < 0$	Inferior (If income increases $\rightarrow$ decrease in quantity demanded)
$0 < \epsilon_{i,I} \leq 1$	Normal (an increase in income increases q. demanded)
$\epsilon_{i,I} > 1$	Superior (Luxury) (an increase in income $\rightarrow$ increases quantity demanded by a lot)
$\epsilon_{i,I} = 0$	Independent (Neutral) to income.

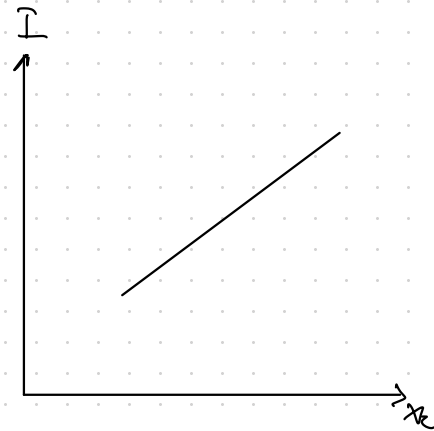
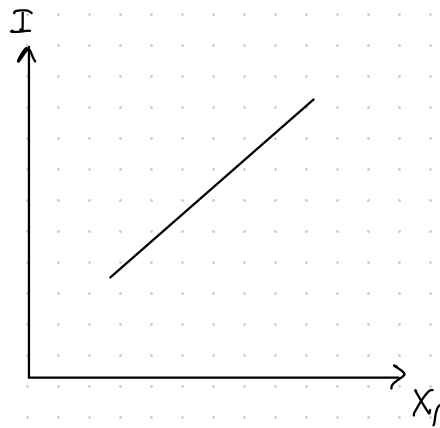
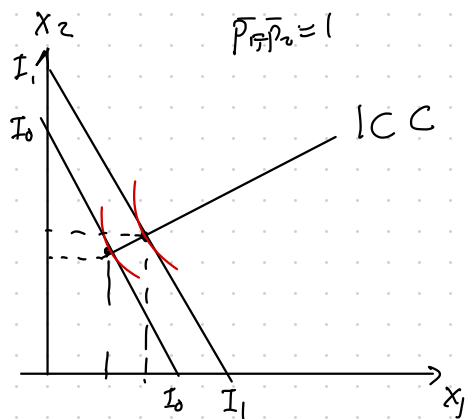
Engel Aggregation :

$$S_i = \frac{P_i X_i}{I} \Rightarrow \frac{P_1 X_1 + P_2 X_2}{I} = \frac{I}{I} \Rightarrow S_1 + S_2 = 1$$

\ /
weights : $0 \leq S_i \leq 1$

The Engel Aggregation : $S_1 \cdot \epsilon_{1,I} + S_2 \cdot \epsilon_{2,I} = 1$

Relationship to Income Elasticities



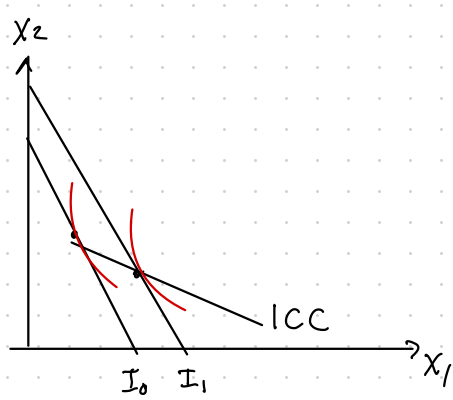
$$\Delta I = I_1 - I_0 > 0 \Rightarrow \Delta \% I > 0$$

$$\Delta x_1 > 0 \Rightarrow \Delta \% x_1 > 0$$

$$\Delta x_2 > 0 \Rightarrow \Delta \% x_2 > 0$$

$S_1 \epsilon_{1,I} + S_2 \epsilon_{2,I} = 1 \Rightarrow$ - One of them needs to be less than 1, the other more than 1.

- $\epsilon_{i,I} = 1$ for both.

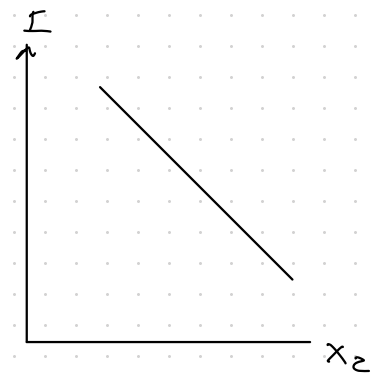
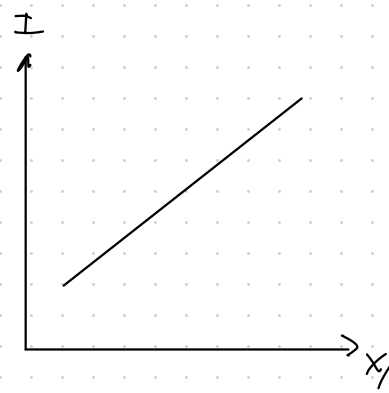


$$\left. \begin{array}{l} \Delta \% I > 0 \\ \Delta \% x_1 > 0 \\ \Delta \% x_2 < 0 \end{array} \right\} \Rightarrow \begin{array}{l} \epsilon_{1,I} > 0 \\ \epsilon_{2,I} < 0 \end{array}$$

$$s_1 \epsilon_{1,I} + s_2 \epsilon_{2,I} = 1$$

$$\Rightarrow \epsilon_{1,I} > 1 \Rightarrow 1 \text{ is superior}$$

$\therefore$ If one good is inferior
the other one must be
superior.



Engel curves

Ex: $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$

$\downarrow$ $MRS = \frac{\alpha}{1-\alpha} \frac{x_2}{x_1}$

$x_1^m = \frac{\alpha I}{p_1}$

$x_2^m = \frac{(1-\alpha)I}{p_2}$

$p_1 = p_2 = 1 \Rightarrow x_1^m = \alpha I$
 $x_2^m = (1-\alpha)I$

$ICC(I) \equiv (\alpha I, (1-\alpha)I)$
 $\swarrow \quad \searrow$
 $x_1 \quad x_2$

$\frac{x_1}{\alpha} = I$

$\frac{x_2}{1-\alpha} = I$

$\frac{x_1}{\alpha} = \frac{x_2}{1-\alpha} \Rightarrow \frac{\alpha \cdot x_2}{1-\alpha \cdot x_1} = 1$

$\downarrow$
 same
 as the

relationship obtained
 by the optimality
 condition.

$U(x_1, x_2) = \min \left\{ \frac{x_1}{\alpha}, \frac{x_2}{\beta} \right\}$



$\frac{x_1}{\alpha} = \frac{x_2}{\beta}$

Location of Kink

$\hookrightarrow x_2 = \frac{\beta}{\alpha} x_1$ This is the
 ICC

